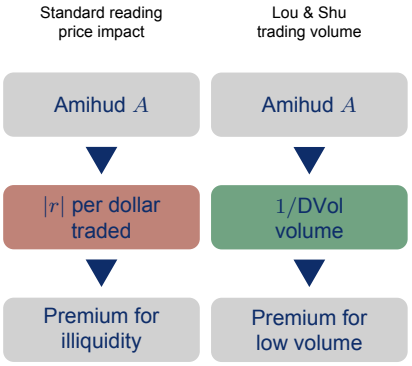
Price Impact or Trading Volume: Why Is the Amihud (2002) Measure Priced?

Lou & Shu (2017), replication on Binance and upgrades. Group 12: Andrey Tsember, Sofia Shapovalova, Ksenia Stepanova, Marat Bineman, Alexia Fox. Detailed notes to the slides; every number is generated from `results/` (src/build_docs.py).

Problem



The Amihud measure is priced, but for what?

Stocks that are illiquid by Amihud earn a premium, usually read as compensation for price impact. Lou & Shu ask whether that reading is right.

$$A_{i,t} = \frac{1}{D} \sum_{d} \frac{|r_{i,d}|}{\text{DVol}_{i,d}}$$

Paper's premium: quintile H-L on A, 0.56 (t 2.36) %/month (NYSE/AMEX 1964–2012, 588 months).

Why it matters:

A is used both to *measure* liquidity and to *explain* returns; these are two different questions.

Source: Literature: Amihud (2002) → Goyenko–Holden–Trzcinka (2009) → Lou & Shu (2017) → Amihud & Noh (2021) → EDGE, Ardia–Guidotti–Kroencke (2024)

Key observation: across assets, A is almost only volume

Dollar volume varies enormously across assets and $|r|$ varies little, so $A \approx A_C$.

0.899 corr(A, A_C), paper

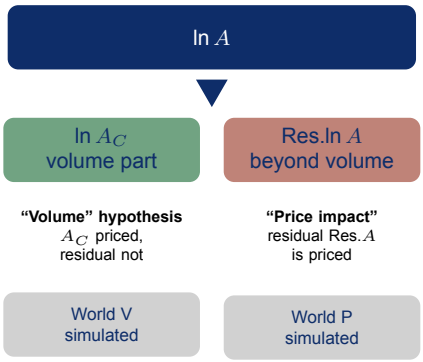
0.956 corr(A, A_C), Binance

	paper (NYSE/AMEX)	us (Binance)
Q75/Q25 of A_C	116.60	9.40
Q75/Q25 of $\|Ret\|$	2.18	1.75
corr(A, $\|Ret\|$)	0.489	−0.133

- Our dispersion of liquidity is an order of magnitude smaller (volume filter and Binance listing policy are a hypothesis for why), but the conclusion is the same: A is close to A_C.
- Difference: for coins, volume rises with volatility, so corr(A, $|Ret|$) is negative.

Source: results/20261016T231216Z_3e8eb87e/paper_vs_us.csv (T1)

Method: a “constant” measure and a residual



If A_C is priced and the part of A beyond A_C is not, the Amihud premium is a volume premium.

$$A_C = \frac{1}{D}\sum_d \frac{1}{\text{DVol}_d}$$

$$\ln A = a_t + b_t \ln A_C + \text{Res.}\ln A$$

First we tested on simulations whether the pipeline tells the hypotheses apart (300 coins × 90 months; 3 of 17 criteria fail):

- world V: A_C significant in 100% of runs, Res.lnA in 4%;
- world P: Res.lnA significant in FM in 100%, in the sort only 76%; ln A_C falsely significant in 18%;
- real SE of the A_C spread is 3.4–3.5× the simulated one, so real power is much lower.

Source: results/simulations/sim_world_summary.csv; results/upgrades/robustness/

Dataset overview and tests

Quintile sorts and Fama–MacBeth, signal from month $t-2$, Newey–West(6) t-statistics; $r_{i,t} - \hat{\beta}_i \,\text{CMKT}_t = \gamma_0 + \gamma_1 X_{i,t-2} + \textit{controls} + \varepsilon_{i,t}$



Binance spot, USDT pairs, 2019-01…2026-09

626 coins in total — **93** months — **265** coins per month — **588** months in the paper (1775 stocks/month)

FM on 87 months with 214 coins; CMKT instead of FF3 (no size, no B/M); later-delisted pairs kept, no delisting returns (D10). Data audit: 8 redenominations adjusted like splits, 2 reused tickers split. Base specification fixed before the first real-data run; config hash matches docs/preregistration/; main run config 3e8eb87e (adds J1, J2).

The paper’s results (Table 3, Table 4)

In the paper the spreads on A and A_C are positive and significant, the spread on Res.A is not; in FM, Res.lnA is significantly negative.

signal	quintiles Q1…Q5, %/month	H-L
A	0.96 / 1.16 / 1.23 / 1.29 / 1.53	0.56 (t 2.36)
A_C	0.96 / 1.13 / 1.23 / 1.28 / 1.57	0.61 (t 2.95)
Res.A	1.39 / 1.29 / 1.19 / 1.10 / 1.21	−0.17 (t −1.05)

FM ln A_C
0.183 (t 4.79)

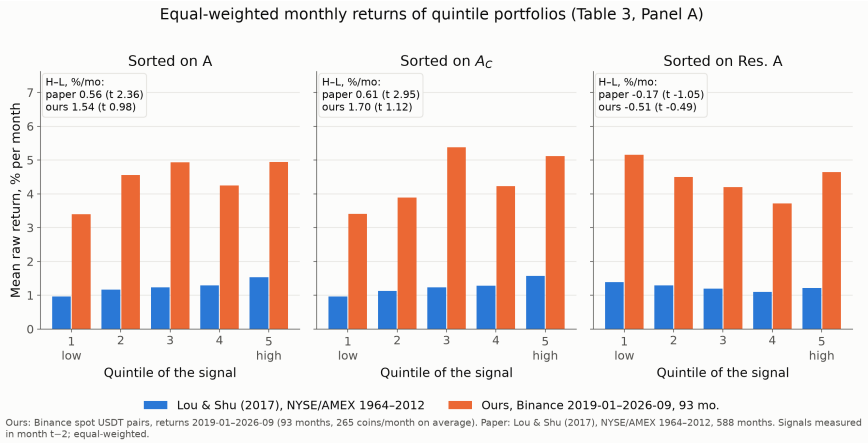
Res.lnA
−0.383 (t −5.07)

The paper's logic: the part of A beyond volume is priced with a negative sign, so the premium is not for price impact.

Source: src/paper_numbers.yaml (transcribed with pdftotext, checked against the PDF by hand; no automated check)

Our replication next to the paper

The sign of H-L matches (Q5 > Q1 for A and A_C, negative for Res.A), but our quintiles are not monotone and our spreads are noisy and insignificant.



H-L A_C: 1.70 (t 1.12)
paper 0.61 (t 2.95)

H-L A: 1.54 (t 0.98)

H-L Res.A: −0.510 (t −0.49)

s.d. of a coin's monthly return: 43.9%.

Without March 2021 (altcoin season) the A_C spread is 0.09 %/month (t 0.08).

Source: results/20261016T231216Z_3e8eb87e/fig_quintiles.png: the paper (Table 3, Panel A) and us on the same axes

The paper versus us

25/28 signs agree

8/21 paper's significant rows repeat

mostly Res.lnA (6 of 7; 5 at the world-0 cut-off, CI 2.04–2.96); none at HLZ.

metric	paper (NYSE/AMEX 1964–2012)	us (Binance)	match
corr(A, A_C)	0.899	0.956	yes
Q75/Q25 of A_C	116.60	9.40	no
corr(A, |Ret|)	0.489	−0.133	no
H-L on A, raw (%/mo)	0.56 (t 2.36)	1.54 (t 0.98)	sign only
H-L on A_C, raw (%/mo)	0.61 (t 2.95)	1.70 (t 1.12)	sign only
H-L on A_C, alpha (%/mo)	0.44 (t 3.20)	0.781 (t 0.62)	sign only
H-L on Res.A, raw (%/mo)	−0.17 (t −1.05)	−0.510 (t −0.49)	yes
FM ln A, model (1)	0.119 (t 3.01)	0.580 (t 1.24)	sign only
FM ln A_C, model (2)	0.183 (t 4.79)	0.760 (t 1.53)	sign only
FM Res.lnA, model (3)	−0.383 (t −5.07)	−5.09 (t −2.46)	yes
FM Res.lnA, model (4) with ln A_C	−0.248 (t −3.14)	−8.10 (t −2.37)	yes
alpha of H-L(A) on IMLA_C (%/mo)	−0.04 (t −0.77)	0.234 (t 0.50)	yes*

* Both insignificant, opposite signs. Also repeated: ln|Ret| in Table 2 (−9.05 (t −2.50)), ln A_C in the first half (1.98 (t 2.59)). World-0 cut-off 2.15–2.16; below it: Res.lnA with IVOL (−6.33 (t −1.98)), raw returns (−3.70 (t −2.11)).

Why the volume premium may be insignificant: (1) power (measured): 93 vs 588 months, MDE80 of the A_C spread 4.27 %/month; untested: (2) another market, where volume rises with volatility; (3) no size or B/M control.

Source: All 28 rows: README.md, paper_vs_us.md

HF benchmark: λ from 5-minute bars

Trade sign from taker volume. In one 36-month window ln λ has a negative FM slope, unlike Table 9 model (1); the A_C premium is absent in this window, so this may be a window effect.

0.615 corr(A, λ), us

0.737 corr(A, λ), paper

FM ln λ: −2.59 (t −4.03)
(size of NW(6) at this T not calibrated)

	paper (T9, T10: λ sample 1983–2012, 358 mo.)	us (Binance 2023-10…2026-09, 5-min bars)
corr(λ, A)	0.737	0.615
corr(λ, A_C)	0.747	0.547
corr(λ, Res.A)	0.149	0.294
FM ln λ, model (1)	−0.034 (t −0.67)	−2.59 (t −4.03)
FM ln λ with ln A_C, model (2)	−0.155 (t −2.51)	−4.30 (t −3.57)
FM ln A_C with ln λ, model (2)	0.18 (t 3.80)	1.06 (t 1.97)
T10: cost part, model (3)	0.015 (t 0.35)	−1.42 (t −3.92)
T10: non-cost part, model (3)	0.191 (t 3.10)	0.957 (t 1.88)

13 708 coin-months; no Lee–Ready. Bar λ vs trade-level λ: corr of ln λ 0.86 (Spearman 0.80) on 69 coin-months (2024-06, 2025-06); bar λ is 8.1× larger, so only logs and ranks compare. Without λ, ln A_C is −0.435 (t −1.56). Post hoc: without exit months ln λ −3.36 (t −2.75); with IVOL −2.11 (t −2.93); with dollar volume −3.92 (t −3.78).

Source: results/upgrades/hf_lambda/ (T9A_corr, T9B_fm, T10_fm, validation_summary); paper Tables 9–10

Liquidity or mispricing?

Sentiment, market illiquidity and January: none gives a clean answer.

1 Sentiment
L−S insignificant (2.57 (t 1.84)); main FM slope 0.586 (t 1.99) passes 1.96 only marginally, not HLZ. 0 of 4 non-post-hoc L−S variants pass 1.96.

2 Market illiquidity
Paper's sign, but weakens with a market control (post hoc). Our AR(2) state uses the full sample (look-ahead) and tracks volatility, so no claim "against liquidity".

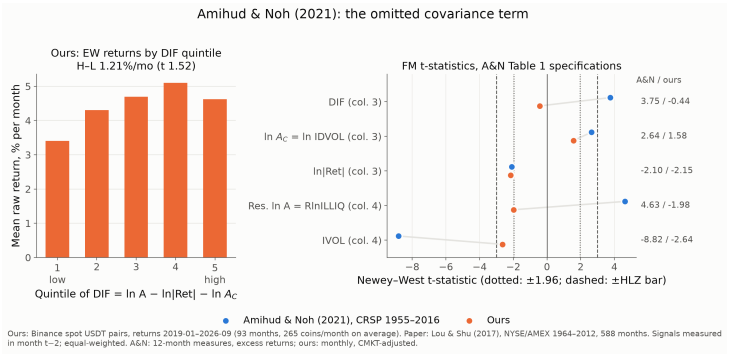
3 January
Only 7 Januaries: a placebo for us. Earnings-announcement test impossible: coins have no earnings.

test	paper (AT_C; BW / PS)	us (A_C; Fear & Greed / AR(2) of A)
Sentiment → L−S (Table 13)	0.24 (t 1.81)	2.57 (t 1.84)
Sentiment → short leg	−0.25 (t −2.67)	−1.37 (t −1.70)
Sentiment → FM slope of ln A_C	—	0.586 (t 1.99)
Market illiquidity → L−S (Table 12)	−0.048 (t −2.38)	−4.30 (t −3.58)
same, + CMKT(t−1), post hoc	—	−3.16 (t −2.60)
January slope (Table 11 (1); paper: ln AT_C)	−0.367 (t −2.27)	January 0.699 (t 0.39), n 7
Other months	0.23 (t 6.28)	0.766 (t 1.62), n 80

Source: results/upgrades/sentiment/paper_vs_us.csv, panel_b.csv; results/20261016T231216Z_3e8eb87e/T7_january.csv

Who is right: Lou & Shu or Amihud & Noh?

The A&N premise holds in crypto, but DIF is insignificant and our power against an A&N-sized effect is unknown: the data do not settle the dispute.



$$\ln A = \ln|Ret| + \ln A_C + \text{DIF}$$

COV < 0 in 98.8% of coin-months

DIF given ln A_C, ln $|Ret|$: −1.64 (t −0.44) (monthly), 14.11 (t 1.55) (12-month, A&N's sign); A&N 0.996 (t 3.75)

fDIF (12-month): 45.12 (t 2.14), the only main-specification result toward A&N, below HLZ (raw returns: monthly DIF 5.43 (t 2.03))

Res.lnA: −6.33 (t −1.98) with IVOL, −0.143 (t −0.05) with ln $|Ret|$ + IVOL; R^2 of Res.lnA on ln $|Ret|$ + DIF = 0.87.

Source: results/upgrades/amihud_noh/; verdict rule (not pre-registered): none / AN_weak / none at minimum coins k+10 (main) / 30 / 50 (sensitivity added post hoc)

Critique

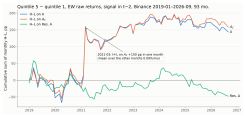
The paper's weak link is the interpretation of the residual; ours is power and one month.

1 Residual = volatility?
The decomposition drops the covariance term (A&N). Our residual is mostly $\ln|Ret|$ (R^2 0.61); IVOL alone does not absorb it (−6.33 (t −1.98)).

2 False dichotomy
"Volume, not liquidity": liquid assets trade more (Amihud–Mendelson 1986).

3 Noisy λ?
Lee–Ready errors could attenuate the paper's λ (untested). Our λ is priced negatively, so "noise hides a premium" is not supported.

4 US before 2012
After 2012 the OpenAP Illiquidity L−S fell (0.324 (t 2.78) → −0.142 (t −1.47)); FF3+UMD alpha insignificant in both (0.136 (t 1.60) → 0.165 (t 1.66)); SMB loading fell (0.705 (t 14.93) → 0.241 (t 4.46)).



Our side: without 2021 the A_C spread is 0.432 (t 0.42). On MOEX the positive $\ln A_C$ slope repeats at 1.96 (model (2) 0.308 (t 2.87); raw returns: (2) 0.114 (t 1.29), (4) 0.100 (t 1.15)), but the paper's negative Res.lnA does not (−0.038 (t −0.13)). OpenAP (NYSE, price > $5, annual rebalancing) is not the paper's design.

Overfitting: could these results be overfitted?

Authors: subperiods, NASDAQ out of sample, annual measures, raw returns. Us:

1 Pre-registration
Base specification fixed before the first run; changes J1, J2 logged. In that first run no T4 slope reached 1.96 (Res.lnA (3) −128.81 (t −1.06)); the J1 audit came after it. Upgrade specs and verdict rules not pre-registered; post hoc variants labelled.

2 Multiple testing
Base family 73 tests, |t| > 1.96 for 16 (3.65–5.7 expected under the null); |t| > 3 for 0; Bonferroni/BH 0/0. World 0: NW(6) gives 6.5%–9.5% false |t| > 1.96; 5% cut-off 2.00–2.22.

3 Stability
Month-block bootstrap: CI excluding zero stable for 2 of 11 statistics. Placebo shuffle (20 perms): |mean t| $\leq$ 0.32. $\ln A_C$ 1.98 (t 2.59) first half, −0.280 (t −1.09) second.

Simulations: 14 of 17 criteria pass (5 under a loose "share ≤ 20%" rule; 11 with 5%); failed: world P, hl_ResA: 76.0% (rule ≥ 80%); world 0, fm3_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%); world 0, fm4_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%). Survivorship: later-delisted pairs kept, no delisting returns. Costs: break-even 1.52% per side, from an insignificant gross spread. Out of sample: MOEX, US after 2012; robustness: without 2021.

Source: results/upgrades/robustness/ (ledger_summary, bootstrap_ci, world0_calibration); results/20261016T231216Z_3e8eb87e/placebo.csv

Next steps: three more months

The priority is power and a proper spread measurement.

1 Market caps
Supply from CoinGecko/CMC → size control, AT/AT_C, VW portfolios.

2 Real spreads
Intraday EDGE on 5-minute bars and order-book snapshots.

3 Better λ
From midquotes, over the full history, not 36 months.

4 A&N test
12-month measures, minimum coins fixed in advance; residual on ln VOLUME (Table 6).

5 Decay
McLean–Pontiff post-publication decay within the US sample.

We need more months or less noise, not more specifications.

MDE80 for FM $\ln A_C$: 1.39 per unit of $\ln A_C$ against the paper's slope 0.183 (7.6× larger); per 1 s.d. 2.15 against 0.46 (4.6×).

Takeaways

- 01 Across assets the Amihud measure is almost only volume: corr(A, A_C) 0.956 on Binance, 0.899 in the paper.
- 02 Signs agree for 25 of 28 rows; the negative Res.lnA slope repeats, the volume premium is not significant.
- 03 Power is the bottleneck: 93 vs 588 months, MDE80 of the A_C spread 4.27 %/month.

Beyond volume, A is not priced positively.

Volume premium in crypto? Not with 93 months.

Definitions

- $A = \text{mean}_d(|r_d|/\text{DVol}_d)$, $A_C = \text{mean}_d(1/\text{DVol}_d)$; Res.lnA is the residual of ln A on ln A_C, estimated each month.
- Res.ln A regression: ln $A = a_t + b_t$ ln A_C + Res. ln A.
- Amihud & Noh: ln A = ln $|Ret|$ + ln A_C + DIF; DIF reflects cov$(|r|, 1/\text{DVol})$.
- λ (Hasbrouck): 5-minute return on $\sum$ sign · $\sqrt{\text{DVol}}$; ours uses SVol $= \sqrt{B} - \sqrt{S}$ from taker volume per bar.
- IMLA$_C$: tercile H-L on A_C; CMKT: EW market of the universe; CMOM: tercile H-L on Ret$[-12,-2]$.

Source: src/measures.py, src/hf_lambda.py, src/factors.py

Simulations: worlds V / P / 0

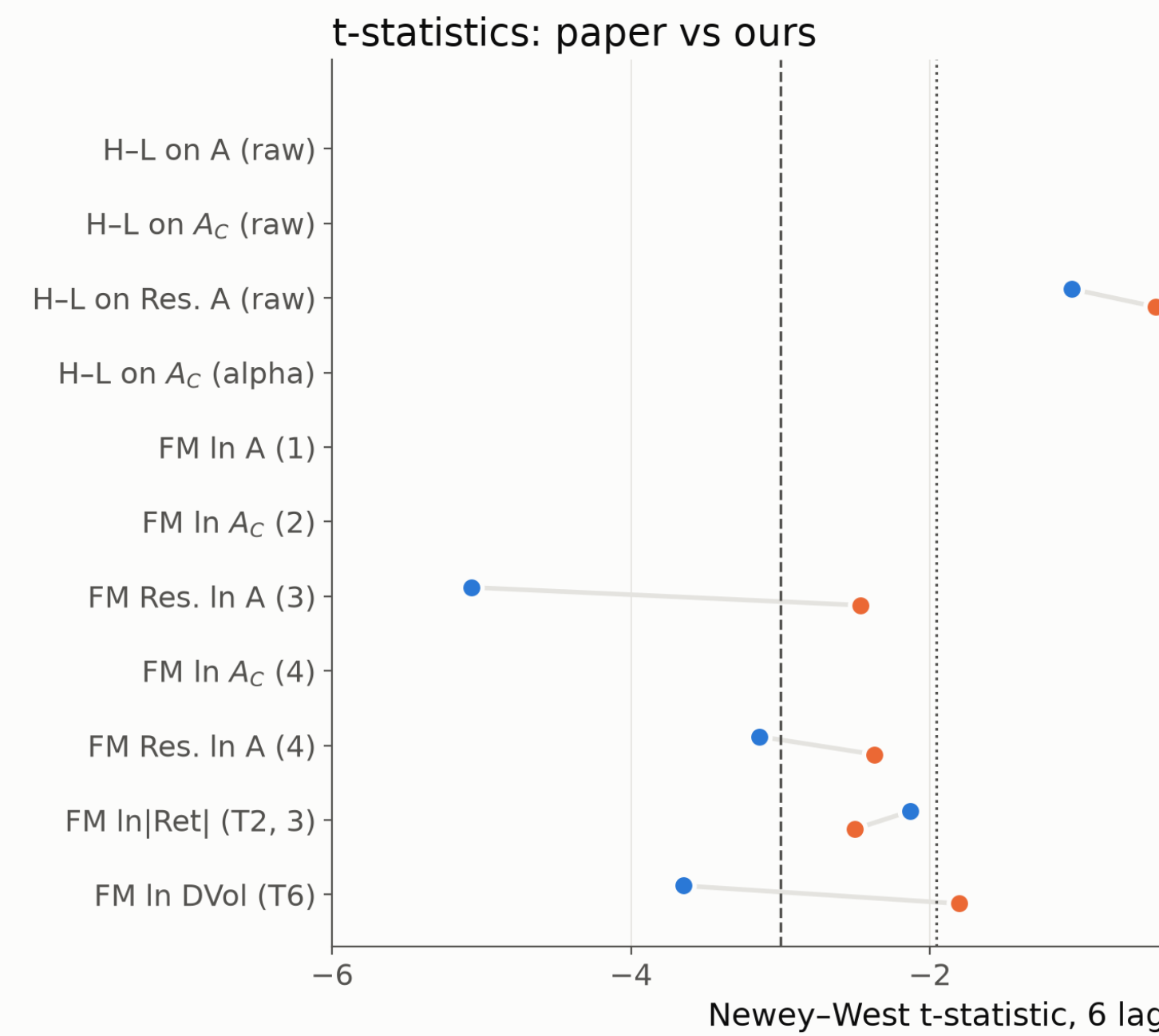
In world V A_C is priced, in world P Res.lnA; in world 0 NW(6) false positives are above 5%, significantly so only for Res.lnA (9.5%; a P0 criterion fails).

world	statistic	expectation	share significant	mean t	replications
V	fm2_lnA_C	priced: high share of t > 1.96	100.0%	7.24	50
V	hl_A_C	priced	100.0%	6.61	50
V	fm3_ResLnA	not priced: share of |t| > 1.96 near 5%	4.0%	0.06	50
V	hl_ResA	not priced	8.0%	−0.05	50
P	fm3_ResLnA	priced	100.0%	4.78	50
P	hl_ResA	priced (the sort is weaker than FM)	76.0%	2.83	50
P	fm2_lnA_C	ideally not priced	18.0%	0.01	50
0	fm2_lnA_C	false positives near 5%	8.0%	−0.03	200
0	fm3_ResLnA	false positives near 5%	9.5%	−0.14	200
0	fm3_ResLnA_nw0	same, without NW lags	5.0%	−0.12	200
0	hl_A_C	false positives near 5%	6.5%	−0.06	200

14 of 17 criteria pass; 5 of them use the loose rule “share ≤ 20%” for unpriced signals (world P falsely prices ln A_C in 18%); with 5% instead, 11 pass. Failed: world P, hl_ResA: 76.0% (rule ≥ 80%); world 0, fm3_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%); world 0, fm4_ResLnA: 9.5% (95% CI 5.8%–14.4% excludes 5%).

Source: results/simulations/sim_world_summary.csv, sim_checks.csv

The paper's t-statistics and ours



Source: results/20261016T231216Z_3e8eb87e/fig_tstats.png (lines at 1.96 and HLZ 3)

Power and robustness

- MDE80: H-L A_C 4.27 %/month against the paper's premium 0.61. FM ln A_C: 1.39 per unit of ln A_C against 0.183; per 1 s.d. (each in its own s.d.) 2.15 (s.d. 1.54) against 0.46 (paper's s.d. 2.53).
- Without 2021: H-L A_C 0.432 (t 0.42), FM ln A_C 0.351 (t 0.97), Res.lnA (3) −4.93 (t −2.54).
- 31 of 45 variants keep the base verdict (same sign, same side of 1.96): 24 of them are insignificant estimates that stay insignificant, and 7 of 8 Res.lnA variants stay significant. Variants: winsorisation, raw returns, weights, subperiods, and 7 post hoc variants without the most influential month. Subperiods: ln A_C 1.98 (t 2.59) and −0.280 (t −1.09).
- Bootstrap: FM Res.lnA (3) [−8.98; −1.52], (4) [−15.29; −2.45]; FM ln A_C (4) borderline (11/20 seeds exclude zero).

Source: results/upgrades/robustness/; results/20261016T231216Z_3e8eb87e/robustness_vs_base.csv

Other upgrades

- **EDGE and costs:** H-L A_C gross 1.65 (t 1.07), net 1.57 (t 1.02); daily EDGE does not measure the spread (gap weight 0.9996); break-even cost 1.52% per side, computed from an insignificant gross spread that rests on March 2021.
- **MOEX TQBR:** FM ln A 0.313 (t 3.01) (at the HLZ bar: FM (1) lnA t 3.01; FM (5) lnA_C t 3.19; FM (5) lnAT_C t 3.04; NW size on MOEX uncalibrated), ln A_C 0.308 (t 2.87), Res.lnA (4) −0.038 (t −0.13) (paper −0.248 (t −3.14)); A_C spread 1.11 (t 1.53), Res.A spread 0.462 (t 1.23) (paper −0.17 (t −1.05)), alpha 0.034 (t 0.13), SMB loading 0.906 (t 8.54); FM with ln ME keeps ln A_C (0.298 (t 2.85); ME from today's share count, a look-ahead). Post hoc: 2023 gives 65% of the total spread.
- **US (OpenAP):** Illiquidity 0.324 (t 2.78) → −0.142 (t −1.47), FF3+UMD alpha 0.136 (t 1.60) → 0.165 (t 1.66); DolVol 0.719 (t 3.98) → −0.013 (t −0.09); t of the raw difference by NW lags 0 / 3 / 6 / 12: −2.58 / −3.00 / −3.08 / −2.97.
- **Test ledger:** base tests and upgrades together, 750 unique tests, |t| > 3 for 66, 14 after Bonferroni: hf_lambda: ln λ 5, T10 cost part 2, lnA_C 1; moex: factor loading 5; sentiment: Long-Short 1. These are factor loadings, 8 rows from the one 36-month λ window (7 λ or cost-part rows, 1 lnA_C row) and one non-main market-illiquidity variant, not distinct findings.

Source: results/upgrades/*/SUMMARY.md; results/upgrades/robustness/ledger.csv

Reproducibility

- `make all` runs data → tests → simulations → base run → report → upgrades → ledger → documents → manifest check; `make slides` builds this deck.
- **make all on a fresh clone** of the final code (results/repro/final_clone/): exit code 0, about 9 min; code tests 172 passed, output-manifest contract 13 passed, 3 xfailed (xfails: historical runs); data/ a directory of the clone (a copy of the cached downloads). Its outputs are the canonical results (9 run manifests); base CSVs vs the previous base run: max abs diff 0. Verdict: reproduced.
- **From-scratch clone** of an earlier version of the code with an empty data/ (every download redone): venv → data → tests → p0 → upgrades-light → upgrade-hf_lambda (3 attempts); hf_lambda needed reruns after network drops; earlier attempts exited 0 despite failed downloads (the code now fails); only the complete last one was compared. Against the current canonical outputs: of 119 result CSVs (timing columns excluded), 0 differ numerically (9 only in text; 2 agree in totals or shared rows); daily panel identical (788 692 rows). Verdict: reproduced.
- Every run writes `run_manifest.json` (sha256 of config, code and data, seed, timings); `make verify` recomputes them from the files, no git needed: config all 10 match, code all 10 match the code here (list: README). Canonical manifests failing the path check: 0; home paths remain only in history files.
- Deviations from the paper: DEVIATIONS.md (D1–D18, U1–U8, J1–J2).

Source: README.md, section "Reproducibility"; results/repro/ and results/repro/final_clone/

What was not done and limitations

- **Reproducibility:** no single clean-clone run both redid the downloads and ran `make all` (`make all` of the final code used cached data; the from-scratch clone ran an earlier version of the code, and its hf_lambda step needed 3 attempts); macOS only.
- **Simulations:** the world-0 size criterion fails for Res.lnA (9.5%); 5 passes use a loose "share ≤ 20%" rule.
- **Pre-registration:** the Res.lnA significance appears only after the J1 audit, designed after the first run; verdict rules and upgrade specifications were not pre-registered; post hoc diagnostics (λ checks, MOEX same-months check, without the most influential month) are labelled.
- **Unevidenced checks:** no stored announcement sources, no saved cache comparison; the number of delisted pairs is not recorded; no delisting returns; the paper's numbers were checked against the PDF by hand, with no automated test.
- **Not done:** market caps (size control, VW by cap); the t+1 leak placebo; corr(AT_C, λ); λ from midquotes.
- **Formalities:** the slides were due 24 hours before the defence (16 October, 13:00), but the deck was first saved on 2026-10-16 19:26 (the first version of slides/deck.tex in our version history): deadline missed. Latin spelling of the names not confirmed by the members; no timed rehearsal recorded.

Source: README.md, section "What was not done and limitations"